

- 3 The growth of the oil-seed crop plant, *Brassica napus*, is affected by high concentrations of salt (sodium chloride) in the soil solution, a condition known as salinity stress.

Methyl jasmonate (MeJA) is a cell-signalling molecule found in plants.

Scientists researched the ability of MeJA to increase the tolerance of seedlings (young plants) of *B. napus* to salinity stress.

- Seedlings were grown in normal conditions for 15 days and then transferred to soils with different concentrations of salt (NaCl), to continue their growth.
- On day 16, the leaves and stems (shoots) of the seedlings were sprayed with $100 \mu\text{mol dm}^{-3}$ MeJA solution.
- On day 44, the seedlings were removed and growth measurements were made.
- Control experiments were carried out to measure the growth of the seedlings in the different salt concentrations with **no** MeJA treatment.

Table 3.1 shows the growth measurements for the roots and the shoots of the seedlings in each of the different salt concentrations.

Table 3.1

treatment		mean root dry mass /g	mean shoot dry mass /g
MeJA / $\mu\text{mol dm}^{-3}$	NaCl /mmol dm ⁻³		
0 (control experiments)	0	0.53	0.24
	110	0.14	0.09
	220	0.04	0.04
	330	0.02	0.02
100	0	0.19	0.12
	110	0.21	0.12
	220	0.13	0.06
	330	0.06	0.04

Table 3.1 shows that the tolerance to salinity stress after spraying with MeJA solution was increased in the roots as well as in the shoots of the seedlings of *B. napus*.

State the conclusions shown by the results in Table 3.1 **and** explain how MeJA sprayed on the leaves and stems of the seedlings can have an effect on growth in root tissue.

.....

.....

.....

.....

.....

.....

.....

.....

.....

[Total: 4]